A Multi-Modeling Approach to the Study of Modeling Complex Bio-behavioral Systems Schank Jeffrey, Psychology, University of California, Davis; May Chris, Psychology, University of California, Davis; Joshi Sanjay, Mechanical & Aeronautical Engineering, University of California, Davis I discuss a multi-modeling approach using computer simulation and robotic models for the study of sensorimotor development in Norway rat pups. I begin by discussing dimensions of modeling in this context and consider the question of whether robotic models are necessary for modeling behavior. Robotic models are a type of simulation model, but do they do anything that computer simulation models do not do? One reason often sited is that many physical variables influence the behavior of a physical system and it is difficult if not practically impossible to represent all of these with adequate laws or rules for physical interactions in a computer simulation. Another, and perhaps more important reason, is that the process of designing, building, and testing robotic models leads to new problems, discoveries, and interpretations precisely because one is working with a physical model. To illustrate this point, I will discuss examples and results of modeling individual and groups of rat pups with robotic models. The Cantor Dust of Conflict Meg Spohn, University of Denver, Denver, CO SCTPLS Annual Conference August, 2005 22 From the classic Cantor Dust, we understand that noise in data transmission tends to scale in a precise pattern. As described by Gleick in his classic book, a common representation of the Cantor Dust is a representation of a transmission with noise in, for example, its middle third, in a scaled iteration down to the bits. Similarly, the intensity of human conflict, as measured by battle deaths over time series, seems to follow scaling patterns not unlike the Cantor Dust, with its own pattern of terrible bursts and terrible silence. War is often described by the soldiers who were there as being composed of short periods of intense violent conflict interspersed with longer periods of quiet tension or tedium. This project examines these patterns, and considers some of the questions that must necessarily accompany them: With a descriptive pattern, what are the risks inherent in forecasting, ethical, philosophical, and otherwise? What are the human and policy implications of potentially using such a pattern, or of allowing it to be used? Nonlinear Hypotheses in Education Research Methodologies Dimitrios Stamovlasis, Education Research Center, Ministry of Education, Athens, Greece The rapid changes of today s world present new challenges on our educational system. Education research dealing with a complex and continuously changing system may not be able to provide the proper support by implementing conventional linear, qualitative or quantitative, approaches. NDS theory and the science of complexity might provide more solid foundation for understanding and decision-making. The present work explores the applicability and the usefulness of Catastrophe Theory for testing nonlinear hypotheses in educational research. Cusp catastrophe models are proposed, which accounts for discontinuities in students' performance. These models implement psychometric variables from neo-Piagetian theories or information processing models, as controls and demonstrate nonlinear interactions between students' mental resources and mental tasks. In addition, cusp catastrophe is tested in others education research areas implementing other behavioral variables such as students attitudes and choices, or teachers selection in workforce recruitment. Measurements were taken at two points in time, and the data analysis involved dynamic difference equations and statistical regression techniques. The nonlinear models proved to be superior to the linear counterparts, and that shows the limitations of statistical modeling used so far in education. Utilizing nonlinear methods and making the paradigm shift realized in behavioral science, seems promising for education and pedagogy. This work demonstrates the feasibility of providing empirical evidences for nonlinear processes in education research, which will build bridges between NDS-theory concepts and pedagogical theories. The Multidimensionality of Health: Tools, One to Estimate the Status of Health and Another One to Evaluate Therapeutic Goals Karl Toifl, Neuropsychiatric clinic, Medical University of Vienna Based on a definition on illness and health, Which includes both individual bio-psycho-social complexity and dynamics of development, two tools will be presented, which are developed to be used in daily clinical practice. The first one should make possible to show the relationship between the estimated levels of the amount of demands a person is confronted with, the effectiveness of strategies to deal with these demands and the actual status of health. The second tool allows in daily clinical practice on one hand to document the multidimensional diagnostic and therapeutic process and on the other hand to evaluate the therapeutic goals. This tool supports the process of permanent qualitative improvement of diagnosis and therapy. SCTPLS 2005 14